Lecture 2: Validation, Overfitting, and Forecast Failure

ECON 417 Business Forecasting

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How Lecture 2 Fits the Course

- Lecture 1: we learned **how to build a baseline forecast**.
- Lecture 2: we learn **when not to trust a forecast**.
- Lecture 3: we will learn **how to stabilize unstable models**.

Key message: Good forecasting is not only fitting. It is controlled skepticism.

Agenda (Today)

1. Why Validation Matters
2. Overfitting vs Underfitting
3. Time-Aware Validation Design
4. Rolling and Expanding Windows
5. Cross-Validation with Time-Series Caveats
6. Bootstrap and Forecast Uncertainty
7. Why Forecasts Fail
8. Teaching Wrap-up

Warm-up Question

Suppose two models are both trained on the same history:

- Model A: very low in-sample error.
- Model B: slightly higher in-sample error.

Question: Which one should we deploy next quarter?

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Answer

We cannot decide from training fit alone. We need out-of-sample validation.

Big Idea of Validation

- A model can look brilliant on old data and fail immediately on new data.
- Validation asks: **Can this model survive new periods?**
- In forecasting, validation must respect the time arrow:

$$\text{Past} \rightarrow \text{Present} \rightarrow \text{Future}$$

Key message: If your evaluation design is wrong, your model choice is wrong.

Intuition First

Overfitting

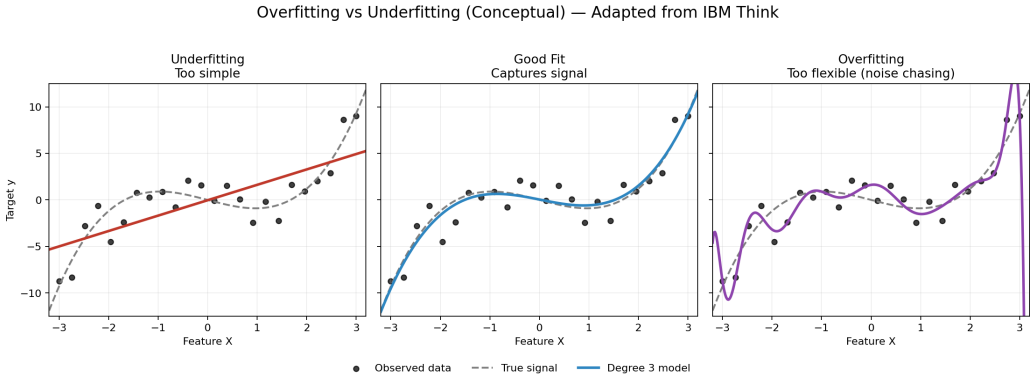
- Memorizes noise.
- Looks great in training.
- Performance drops on new data.

Underfitting

- Too rigid.
- Misses real signal.
- Both train and test errors are high.

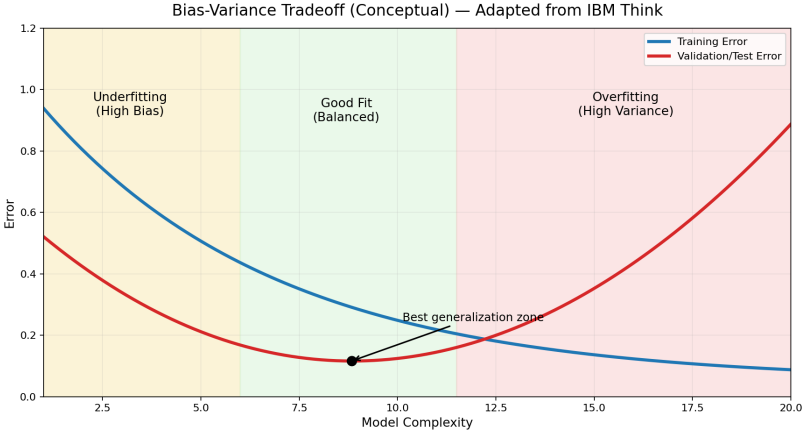
Key message: Our goal is not minimum training error; it is reliable future error.

Underfit vs Good Fit vs Overfit



graph from: IBM

Bias-Variance Tradeoff



graph from: IBM

Fit vs Generalization Summary

- **Underfitting**: model is too simple, so it misses real signal.
- **Overfitting**: model is too tailored to training data, including noise.
- **Good fit**: model captures stable pattern and transfers to unseen data.
- **Practical takeaway**: evaluate on future-like data, not only training fit.

Key message: Generalization is the target; training perfection is not.

Diagnostic with RMSE

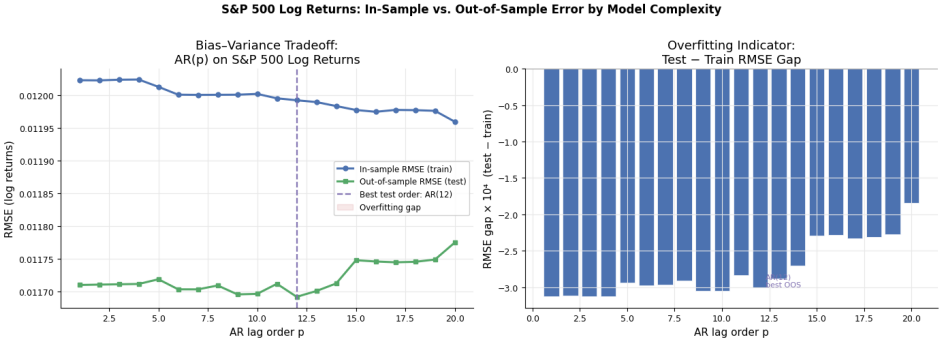
$$\text{RMSE}_{\text{train}} = \sqrt{\frac{1}{n_{tr}} \sum (y_t - \hat{y}_t)^2}$$

$$\text{RMSE}_{\text{test}} = \sqrt{\frac{1}{n_{te}} \sum (y_t - \hat{y}_t)^2}$$

- If complexity increases and $\text{RMSE}_{\text{train}}$ keeps falling, this is normal.
- If $\text{RMSE}_{\text{test}}$ starts rising, overfitting begins.
- Useful warning sign:

$$\text{Gap} = \text{RMSE}_{\text{test}} - \text{RMSE}_{\text{train}} \uparrow$$

Notebook Example: AR(p) on S&P 500 Returns



- As p grows, train error usually improves.
- Test error improves first, then worsens.

Key message: More parameters can mean less forecasting power.

Everyday Analogy

Student analogy

Memorizing past exam answers $\neq$ understanding the course.

- Overfitting model = student who memorizes specific questions.
- Robust model = student who understands core logic.

Class prompt: What kind of model would you hire for an unknown future?

Train / Validation / Test Split

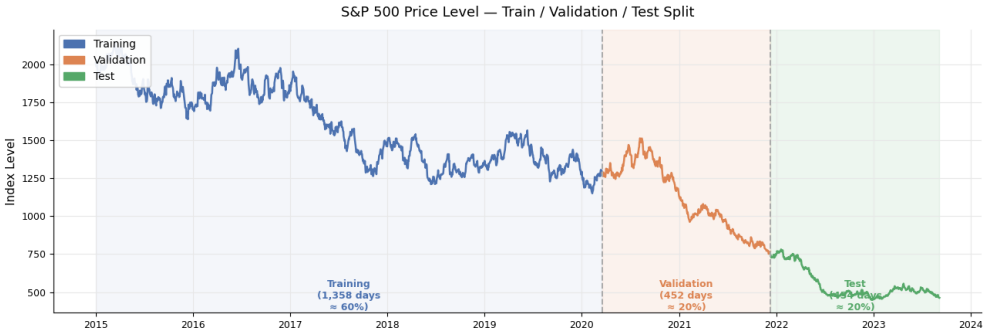
Train Set	**Validation Set**	**Test Set**
Fit candidate models	Tune model and hyperparameters	Final untouched report

Past $\rightarrow$ Model selection $\rightarrow$ Final check $\rightarrow$ Future

- Datasets are typically split into train, validation, and test sets.
- Use this sequence to estimate, tune, and then report a final untouched result.

graph adapted from: IBM

Notebook Example: Chronological Split



- Train set: estimate candidate models.
- Validation set: choose model/hyperparameters.
- Test set: untouched final report.

The Gold Rule: Protect the Test Set

- Do not repeatedly tune based on test performance.
- If test data guides your choices, it becomes another validation set.
- Then your "final" score is optimistic.

Practical rule

Touch test set once, at the end.

Data Leakage: Why It Is Dangerous

- Leakage means future information sneaks into training.
- Result: model appears powerful but cannot be deployed honestly.

One invalid example

Using y_{t+1} to predict y_t in real-time forecasting.

Key message: If you could not know it at forecast time, you cannot use it.

Data Leakage Types

Target Leakage
Uses future-only information
not available at prediction time

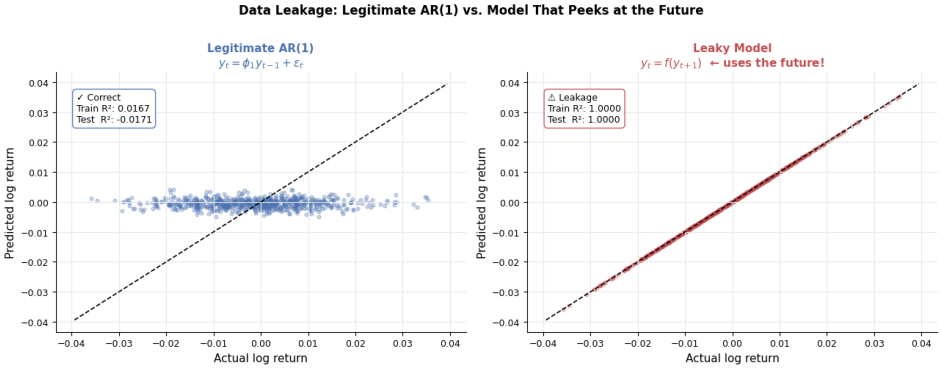
Train-Test Contamination
Test information leaks through
preprocessing or split design

Leakage $\Rightarrow$ Inflated validation score $\Rightarrow$ Deployment failure

- Target leakage and train-test contamination are two key failure patterns.
- Both make a model look better during development than in real forecasting use.

concept adapted from: IBM

Notebook Example: Legitimate vs Leaky Model



- Leaky specification gives misleadingly strong fit.
- Real forecasting performance will collapse.

Two Repeated Validation Schemes

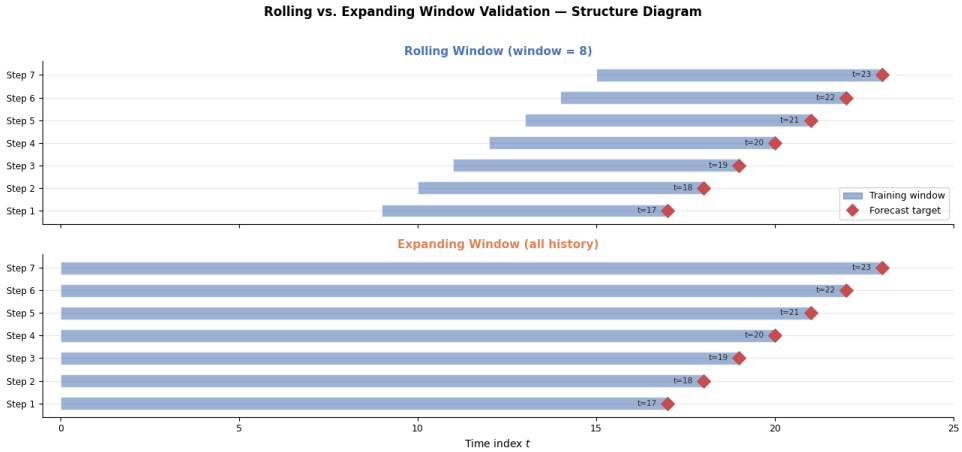
Rolling window

- Fixed history length.
- Focuses on recent regime.
- Faster adaptation.

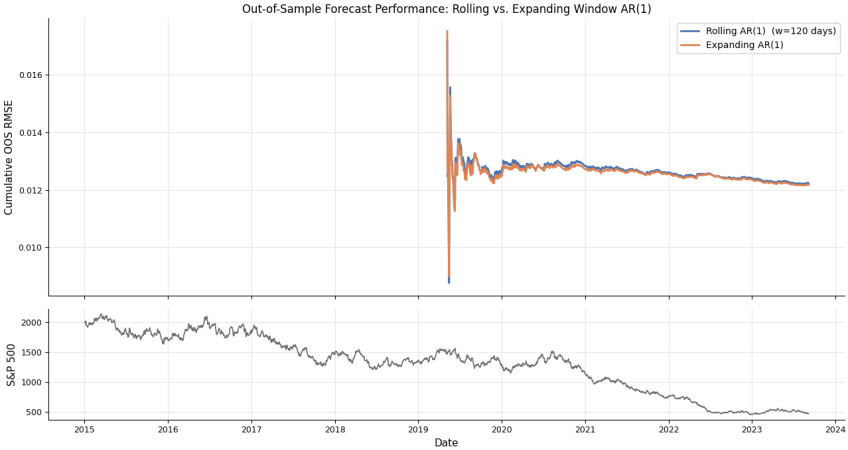
Expanding window

- History keeps growing.
- Lower variance.
- Better if process is stable.

Window Mechanics (Visual)



Empirical Comparison from Notebook



- No method dominates in all periods.
- Method choice depends on instability level.

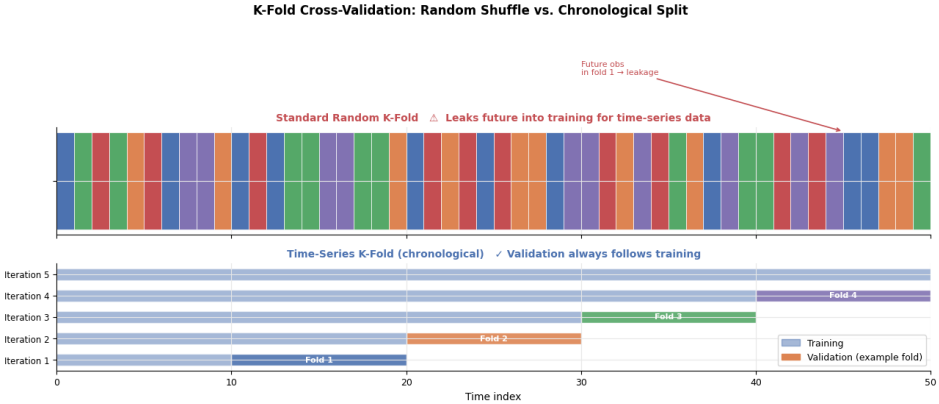
Why Random K-fold Is Risky Here

- Random K-fold assumes observations are exchangeable.
- Time series are ordered and dependent.
- Random shuffling can create hidden look-ahead bias.

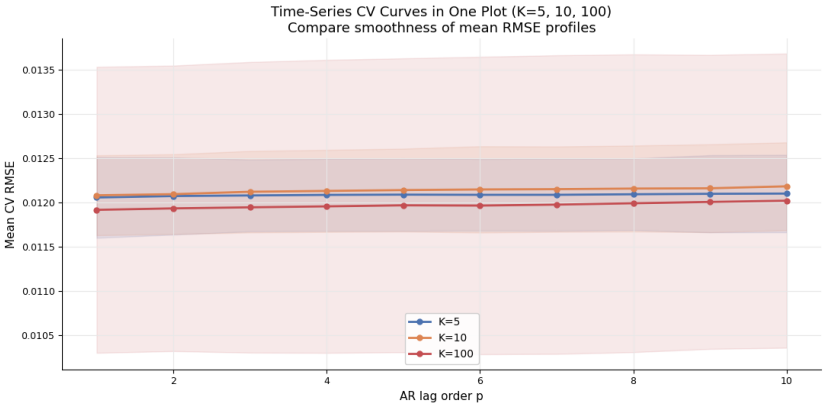
Safe principle

Each validation fold must be later in time than its training fold.

Standard K-fold vs Time-Series K-fold



Fold Count Tradeoff (K=5,10,100)



- More folds can smooth averages but may increase compute and variance sensitivity.

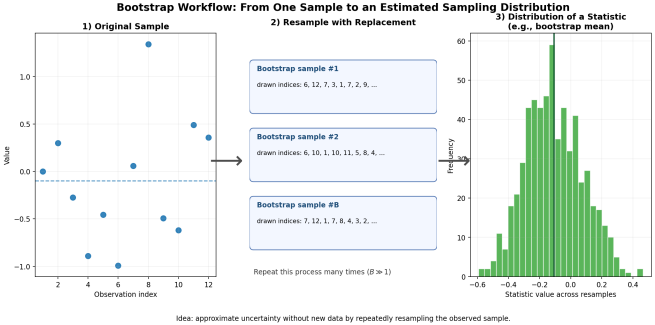
What Is Bootstrapping?

Official definition (NIST/SEMATECH, paraphrased)

Estimate uncertainty by repeatedly drawing subsamples with replacement, recomputing the statistic, and using those values to approximate its sampling distribution.

- Keep sample size fixed.
- Repeat many times (B large).
- Use the empirical distribution for SE/intervals.

Bridge to analogy: each resample is one "shoelace pull."

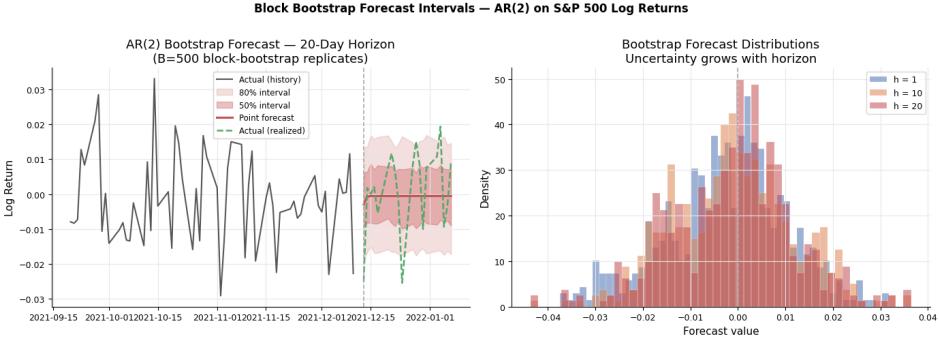


definition adapted from: NIST/SEMATECH e-Handbook

Why We Need Intervals, Not Just Points

- Decision-makers need uncertainty bounds, not one number.
- Block bootstrap preserves local time dependence.
- Output: predictive fan chart across forecast horizons.

Notebook Example: Block Bootstrap Intervals



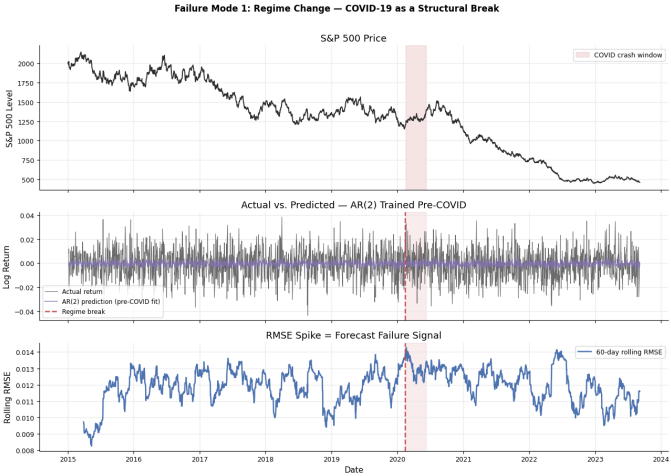
- Bands widen with horizon.
- This is not failure; this is honest uncertainty.

Three Common Failure Modes

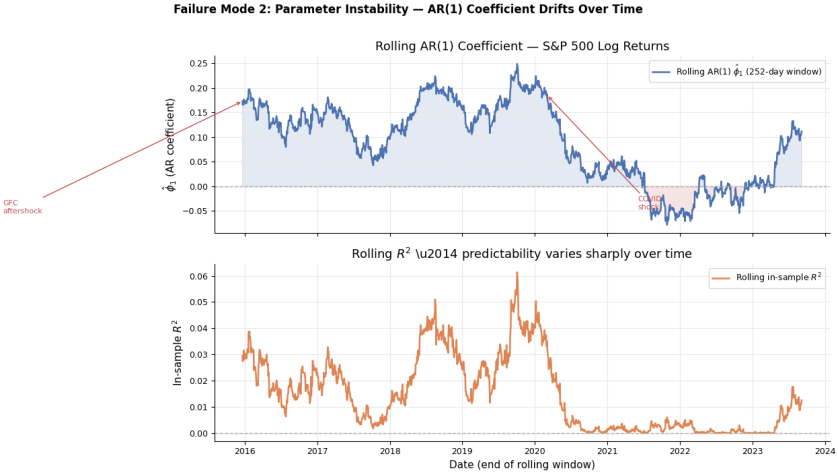
1. **Regime change**: the world shifts (policy, crisis, shock).
2. **Parameter drift**: relationships slowly move over time.
3. **Extrapolation**: future inputs leave historical support.

Key message: A model can be statistically correct and still operationally fragile.

Failure Mode 1: Regime Change



Failure Mode 2: Parameter Instability



Failure Mode 3: Extrapolation Risk

Decision Checklist (Can I Trust This Forecast?)

Before deployment, ask:

1. Is the validation strictly time-aware?
2. Is there a large train-test performance gap?
3. Did we test robustness across windows/folds?
4. Do we have interval uncertainty, not only point forecasts?
5. What failure mode is most likely next period?

In-Class Discussion Prompts

- Which failure mode is most relevant in financial markets?
- If you must choose between lower RMSE and higher stability, what do you choose and why?
- What is one realistic leakage risk in a company forecasting pipeline?

Exit Ticket (3 Sentences)

1. Define overfitting in your own words.
2. Explain why random K-fold can be misleading for time series.
3. Name one reason a good model may fail next year.

End.